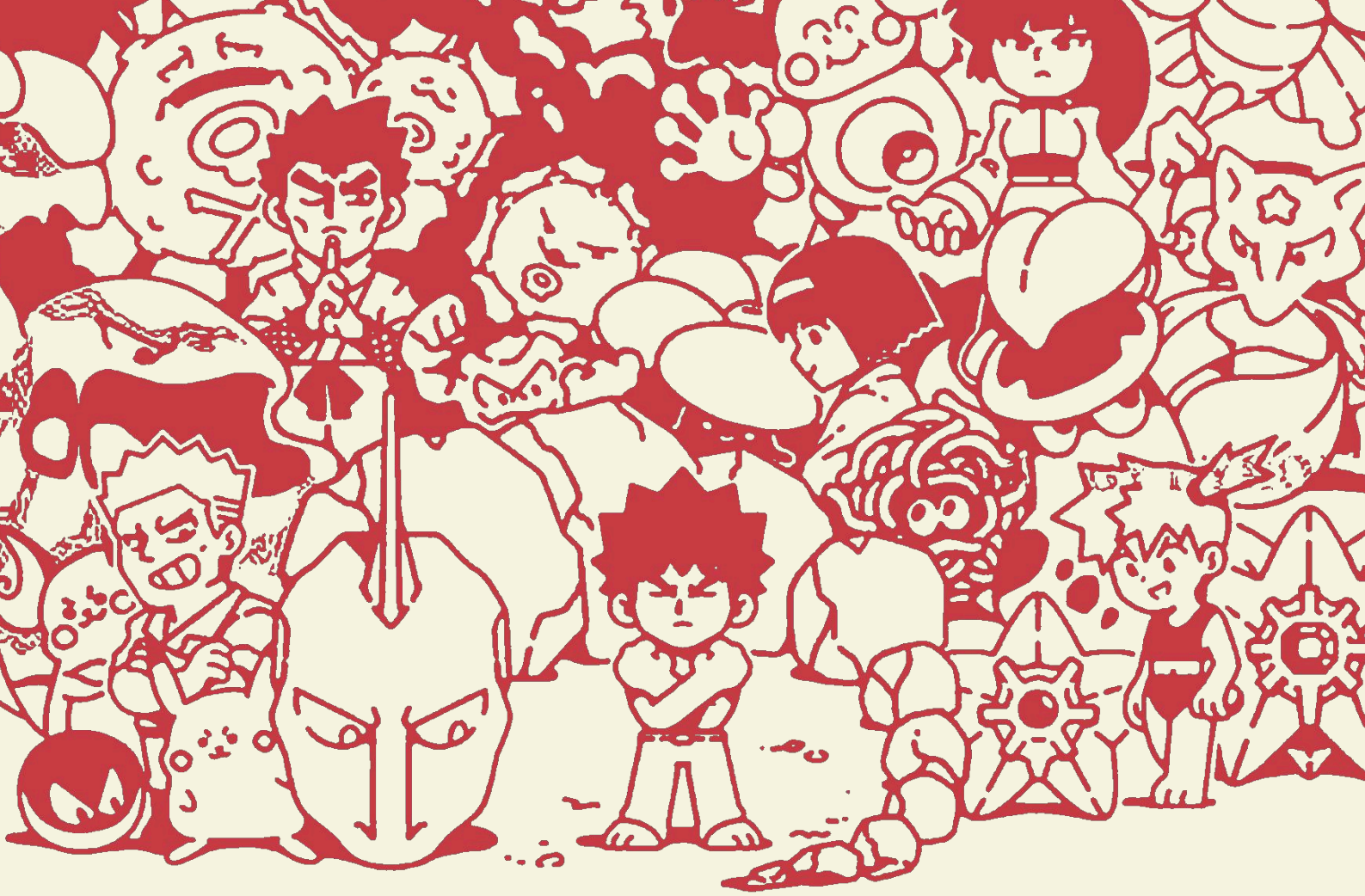
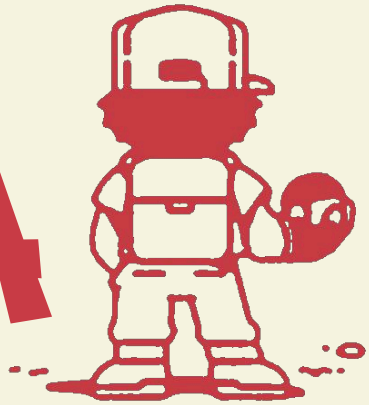




GOTTA CODE EM ALL

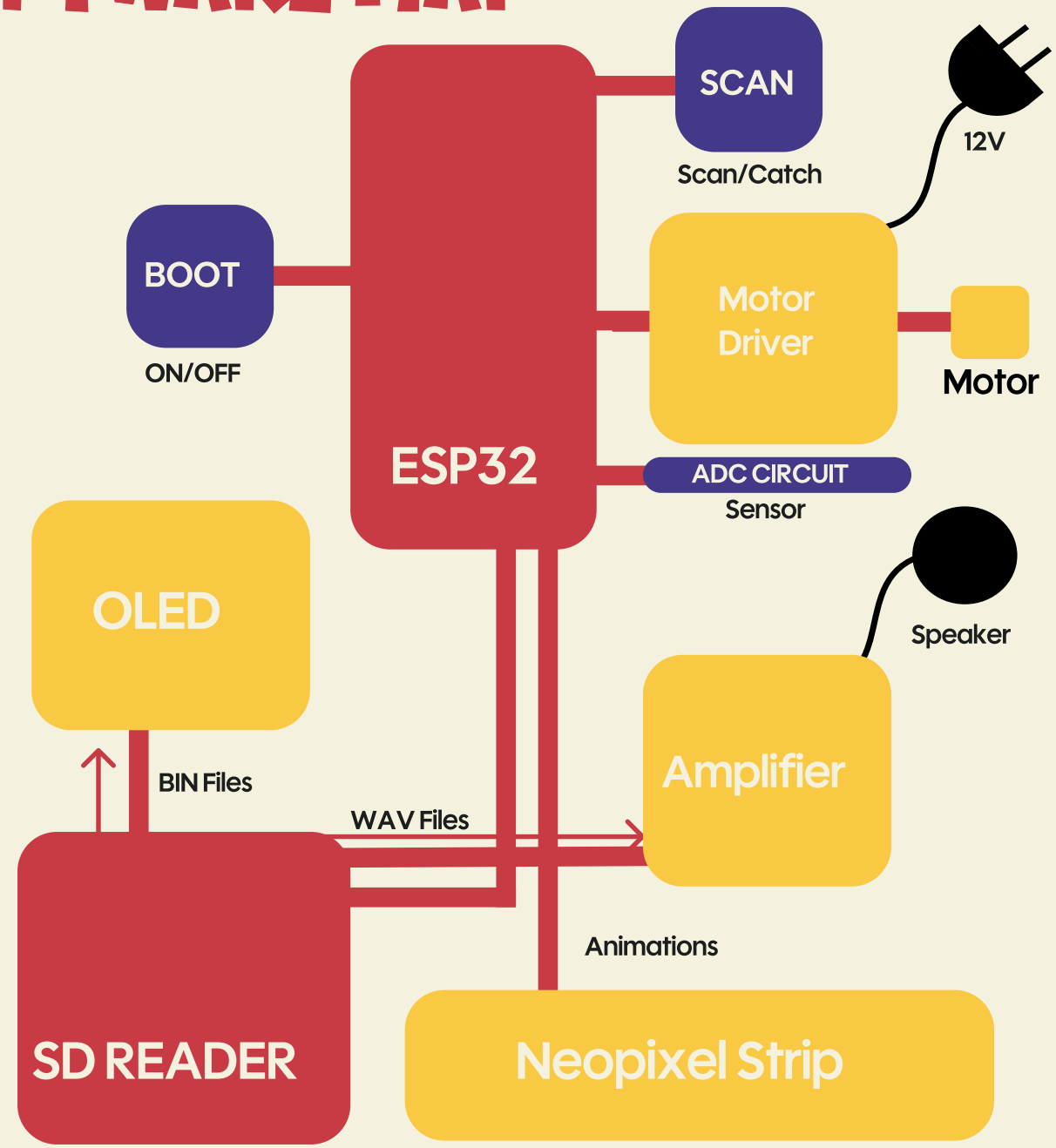
ODT CAPSTONE PROJECT

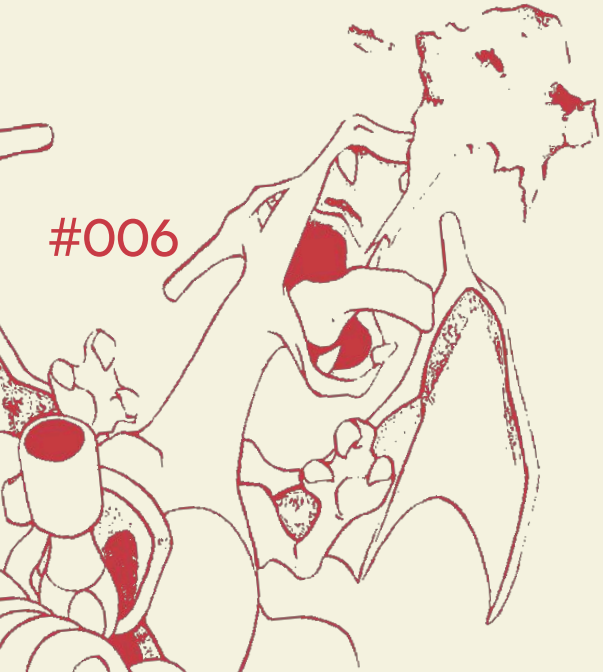
Software Details

Sachit Patnaik
Tanay Dhavate

BITS Design
2025-29

SOFTWARE MAP



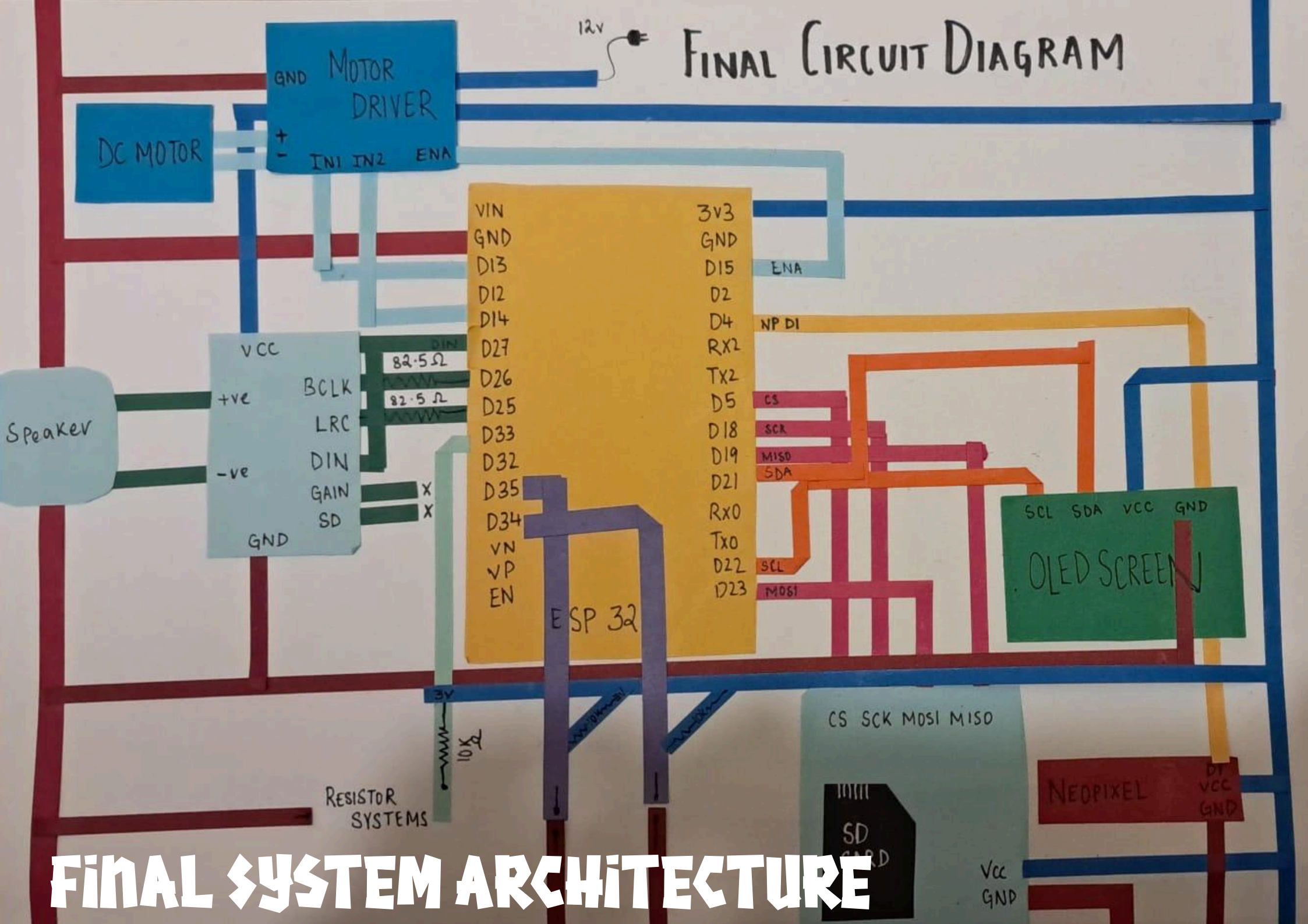


What does the software do?

- Runs the entire Pokédex experience end to end boot sequence, Pokémon identification, animation and audio playback, catch mechanic, and result display
- Manages all hardware simultaneously: OLED display, NeoPixel LEDs, I2S audio, DC motor, ADC reader, and SD card
- Handles all timing, state transitions, memory management, and error recovery on a single thread

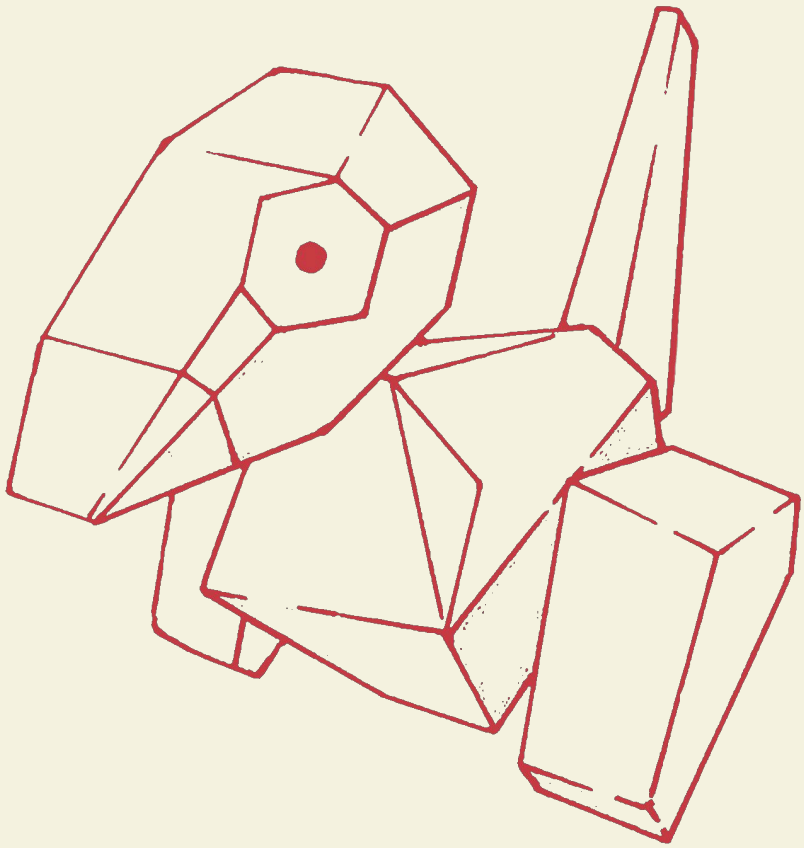
Platform used:

- MicroPython on ESP32


FINAL CIRCUIT DIAGRAM
12V
MOTOR DRIVER
GND
+
-
IN1 IN2 ENA
DC MOTOR
VIN
GND
D13
D12
D14
D27
D26
D25
D33
D32
D35
D34
VN
VP
EN
3V3
GND
D15
D2
D4
RX2
TX2
D5
D18
D19
D21
RX0
TX0
D22
D23
ENA
NP DI
CS
SCK
MISO
SDA
SCL
MOSI
ESP 32
VCC
+ve
-ve
BCLK
LRC
DIN
GAIN
SD
GND
Speaker
82.5Ω
82.5Ω
X
X
3V
10KΩ
RESISTOR SYSTEMS
SCL SDA VCC GND
OLED SCREEN
CS SCK MOSI MISO
SD CARD
NEOPIXEL
DI
VCC
GND
Vcc
GND
FINAL SYSTEM ARCHITECTURE

SYSTEM LOGIC

1. User presses BOOT button
2. ESP32 initialises every peripheral one by one SD card, LEDs, OLED, audio, ADC, buttons, motor
3. Boot animation plays on LEDs and OLED; device reaches ready state
4. User plugs in a Pokémon figure and presses SCAN
5. ESP32 reads the ADC pin 20 times and averages the voltage
6. Averaged value is matched to a Pokémon from the identification table
7. Animation frames and WAV audio file are loaded from the SD card
8. Audio streams via I2S; OLED advances one animation frame every 6 audio chunks, keeping them in sync
9. NeoPixels light up in the Pokémon's theme colour
10. Catch screen appears, user holds SCAN for 5 seconds
11. LEDs fill as a progress bar; countdown ticks on OLED
12. Catch outcome is decided silently using a random roll weighted by rarity
13. DC motor fires three times at set intervals, physically shaking the Pokéball
14. 5 seconds complete, caught or escaped sound plays
15. LEDs flood green or red; OLED displays the result with Pokémon sprite
16. Memory is cleared, state resets, device loops back to ready screen



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